

A Sharp Upper Bound for Random Euclidean Sections of ℓ_p^n in the Logarithmic Regime

Candidate full resolution for arXiv:1510.07284
Run `fa_banach_001`, lane 0

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Status. Candidate full solution, likely valid, pending specialist human review. For every $p \geq C_0 \log n$, the sharp randomized-cube upper bound

$$k \leq C \frac{\varepsilon \log n}{\log(1/\varepsilon)}$$

holds for $(1 + \varepsilon)$ -spherical Haar-random sections of ℓ_p^n . This closes the logarithmic- p , small-distortion wedge left open in Section 6, item 5 of the source paper.

1 Source question

Paouris, Valettas and Zinn study the largest dimension of a Haar-random subspace of ℓ_p^n on which the norm is $(1 + \varepsilon)$ -spherical [1]. In Section 6, item 5, they ask whether the factor $\log(1/\varepsilon)$ in the denominator of

$$k \lesssim \frac{\varepsilon \log n}{\log(1/\varepsilon)}$$

is necessary throughout the range $p \gtrsim \log n$. Their Proposition 6.2 proves the upper bound for $p > (\log n)^2$.

5. The existence of $\log(1/\varepsilon)$ as $p \rightarrow \infty$. Note that Theorem 4.9 and furthermore Corollary 4.10 suggest that the concentration of the ℓ_p norm with $p \gtrsim \log n$ is similar with the one we get for the ℓ_∞ norm. This means that the classical net argument yields random subspaces which are $(1 + \varepsilon)$ -spherical as long as $k \lesssim \varepsilon \log n / \log \frac{1}{\varepsilon}$. We do not know if this $\log(1/\varepsilon)$ term is needed, for this range of p . As an easy corollary of the main result of [34], we have:

Proposition 6.2. *Let $p > (\log n)^2$ and $\varepsilon \in (0, 1/3)$. If the random k -dimensional subspace of ℓ_p^n is $(1 + \varepsilon)$ -spherical with probability greater than $3/4$, then $k \leq C \varepsilon \log n / \log \frac{1}{\varepsilon}$, where $C > 0$ is an absolute constant.*

Figure 1: The open question and Proposition 6.2 in arXiv:1510.07284, PDF page 39.

The theorem below gives the requested upper bound for all $p \geq C_0 \log n$, with absolute constants.

2 Definitions

For a subspace $E \subset \mathbb{R}^n$ define the spherical distortion

$$D_p(E) := \frac{\sup_{x \in E \cap S^{n-1}} \|x\|_p}{\inf_{x \in E \cap S^{n-1}} \|x\|_p}.$$

Thus E is $(1 + \varepsilon)$ -spherical in ℓ_p^n if and only if $D_p(E) \leq 1 + \varepsilon$. All logarithms are natural.

3 Proof intuition

The deterministic comparison $\|x\|_p \leq n^{1/p} \|x\|_\infty$ loses a constant when $p \asymp \log n$, so it cannot resolve very small distortions. The proof replaces this worst-case comparison by a probabilistic soft-maximum comparison at the sharp Gaussian extreme-value level.

Write a Haar-random subspace as the image of an $n \times d$ Gaussian matrix A , whose rows are $G_1, \dots, G_n \in \mathbb{R}^d$. Let u point in the direction of the longest row. Then $\|Au\|_p$ is at least $\max_i \|G_i\|_2$, the maximum of n independent χ_d variables. Choose $v \perp u$. After conditioning on the longest row, the remaining projections $|\langle G_j, v \rangle|$ are stochastically dominated by independent $|N(0, 1)|$ variables. When $p \geq C_0 \log n$, their ℓ_p norm lies, with fixed positive probability, below the same threshold b_n for which $n\mathbb{P}\{|N(0, 1)| > b_n\} = 1/2$.

The χ_d extreme exceeds b_n by the relative amount

$$c \frac{d \log(e \log n / d)}{\log n}.$$

Gaussian singular-value control shows that passing from coefficient-space directions to the Euclidean sphere of $\mathbb{A}\mathbb{R}^d$ costs much less than this gap. Inverting the gap gives exactly $d \lesssim \varepsilon \log n / \log(1/\varepsilon)$. This is the same χ -amplification behind the sharp cube obstruction, but the conditional orthogonal direction makes it work for the finite- p soft maximum without an $n^{1/p}$ loss.

4 Three probabilistic lemmas

Let Z be standard Gaussian and, for $n \geq 2$, let $b_n > 0$ be defined by

$$\mathbb{P}\{|Z| > b_n\} = \frac{1}{2n}. \quad (1)$$

If $L = \log n$, then

$$b_n^2 = 2L - \log L + O(1). \quad (2)$$

Lemma 1 (A Gaussian soft maximum). *There are absolute constants C_0, n_0 such that, whenever $n \geq n_0$ and $p \geq C_0 \log n$, independent standard Gaussians $Z_1, \dots, Z_m$, with $m \leq n$, satisfy*

$$\mathbb{P}\left\{\left(\sum_{j=1}^m |Z_j|^p\right)^{1/p} \leq b_n\right\} \geq \frac{1}{2}.$$

Proof. Put $b = b_n$ and

$$S = \sum_{j=1}^m |Z_j|^p \mathbf{1}_{\{|Z_j| \leq b\}}.$$

For $p > b^2$, the inequalities $\log s \leq -(1-s)$ and $(1-s^2)/2 \leq 1-s$, $0 < s < 1$, give

$$\begin{aligned} \frac{\mathbb{E}S}{b^p} &\leq 2nb \int_0^1 s^p \frac{e^{-b^2 s^2/2}}{\sqrt{2\pi}} ds \\ &\leq 2nb \frac{e^{-b^2/2}}{\sqrt{2\pi}} \int_0^1 e^{-(p-b^2)(1-s)} ds \\ &\leq \frac{2nb\varphi(b)}{p-b^2}, \end{aligned}$$

where φ is the standard Gaussian density. The lower Mills bound $\mathbb{P}\{|Z| > b\} \geq 2\varphi(b)b/(b^2+1)$ and (1) imply

$$\frac{\mathbb{E}S}{b^p} \leq \frac{b^2+1}{2(p-b^2)}. \quad (3)$$

Also

$$\mathbb{P}\left\{\max_{j \leq m} |Z_j| \leq b\right\} \geq \left(1 - \frac{1}{2n}\right)^n.$$

By (2), choosing C_0 sufficiently large makes the right-hand side of (3) smaller than $1/32$. Markov's inequality and $(1 - 1/(2n))^n > 17/32$ for $n \geq n_0$ now yield the claim. $\square$

Lemma 2 (χ -amplification). *There are absolute constants $c_0, c_1 > 0$ and n_0 such that the following holds. Let $L = \log n$, $n \geq n_0$, and $2 \leq d \leq c_0 L$. If $R_1, \dots, R_n$ are independent χ_d variables and*

$$a_d := \frac{d \log(eL/d)}{L},$$

then

$$\mathbb{P}\left\{\max_{i \leq n} R_i > b_n(1 + c_1 a_d)\right\} > \frac{9}{10}.$$

Proof. We include the tail calculation because its d -dependence is the source of the logarithm in the theorem. Put $b = b_n$ and $r = b(1 + c_1 a_d)$. Choose c_0, c_1 small enough that $r^2 \geq 2d$ and $c_1 a_d \leq 1/4$. The χ_d density is

$$f_d(t) = \frac{2^{1-d/2}}{\Gamma(d/2)} t^{d-1} e^{-t^2/2}.$$

Integrating on $[r, r + 1/r]$ gives

$$\mathbb{P}\{R_1 > r\} \geq c \frac{2^{1-d/2}}{\Gamma(d/2)} r^{d-2} e^{-r^2/2}. \quad (4)$$

Indeed, on that interval the Gaussian factor changes by only an absolute factor, while the power t^{d-1} is nondecreasing.

The upper Mills bound and (1) give $e^{-b^2/2} \geq cb/n$. Combining this with (4) and the upper Stirling bound for $\Gamma(d/2)$ yields

$$n\mathbb{P}\{R_1 > r\} \geq c \frac{b\sqrt{d}}{r^2} \left(\frac{er^2}{d}\right)^{d/2} \exp\left(-\frac{r^2 - b^2}{2}\right). \quad (5)$$

Since $b^2 \asymp L$, $r \asymp b$, and $r^2 - b^2 \leq Cc_1 d \log(eL/d)$, taking c_1 sufficiently small in (5) gives

$$\log(n\mathbb{P}\{R_1 > r\}) \geq c'(d-1) \log(eL/d) - C'.$$

The right-hand side tends to infinity uniformly for $2 \leq d \leq c_0 L$; its smallest growth occurs at $d = 2$ and is a positive multiple of $\log L$. Enlarging n_0 therefore makes $n\mathbb{P}\{R_1 > r\} > \log 10$, and independence completes the proof. $\square$

Lemma 3 (Conditioning below the longest row). *Let $G_1, \dots, G_n$ be independent standard Gaussian vectors in $\mathbb{R}^d$, $d \geq 2$, and let i_* be the almost surely unique index maximizing $\|G_i\|_2$. Put $u = G_{i_*}/\|G_{i_*}\|_2$ and choose a measurable $v \in S^{d-1} \cap u^\perp$. Conditional on $(i_*, \|G_{i_*}\|_2, u)$, the random variables $|\langle G_j, v \rangle|$, $j \neq i_*$, are independent and each is stochastically dominated by $|Z|$.*

Proof. Conditional on the longest radius $R = \|G_{i_*}\|_2$, its index and its direction, the remaining rows are independent standard Gaussian vectors conditioned to lie in RB_2^d . Rotational invariance lets us fix the direction v . Write a standard Gaussian vector as (X, Y) along $\mathbb{R}v \oplus v^\perp$. Given $|X| = t$, the probability of the conditioning event is

$$w(t) = \mathbb{P}\{\|Y\|_2 \leq \sqrt{R^2 - t^2}\},$$

with $w(t) = 0$ for $t > R$. The function w is nonincreasing. Weighting the law of $|X|$ by a nonincreasing function gives a stochastically smaller law: for every increasing indicator $\mathbf{1}_{\{t > s\}}$, the elementary opposite-monotonicity covariance inequality gives

$$\frac{\mathbb{E}[\mathbf{1}_{\{|X| > s\}} w(|X|)]}{\mathbb{E}w(|X|)} \leq \mathbb{P}\{|X| > s\}.$$

This proves the one-dimensional domination. Conditional independence of the remaining rows proves the joint assertion. $\square$

5 A direct obstruction in a fixed dimension

Proposition 4. *There are absolute constants $C_0, c_0, \gamma > 0$ and n_0 such that, for $n \geq n_0$, $p \geq C_0 \log n$, and $2 \leq d \leq c_0 \log n$, a Haar-random d -dimensional subspace E of $\mathbb{R}^n$ satisfies*

$$\mathbb{P}\left\{D_p(E) > 1 + \gamma \frac{d \log(e \log n/d)}{\log n}\right\} > \frac{1}{4}.$$

Proof. Let A be an $n \times d$ matrix with independent standard Gaussian entries and rows $G_1, \dots, G_n$. Its image is Haar distributed on the Grassmannian. Let u, v be the two orthogonal directions from Lemma 3. On the event in Lemma 2,

$$\|Au\|_p \geq \|G_{i_*}\|_2 > b_n(1 + c_1 a_d). \quad (6)$$

The i_* coordinate of Av is zero. Lemmas 3 and 1 therefore imply, conditionally on the longest row,

$$\mathbb{P}\{\|Av\|_p \leq b_n\} \geq \frac{1}{2}. \quad (7)$$

Consequently (6) and (7) hold simultaneously with probability greater than 9/20.

We also use the standard Gaussian singular-value estimate [4, Theorem 4.6.1]. Uniformly for the present range of d , with probability tending to one,

$$(1 - \tau)\sqrt{n} \|x\|_2 \leq \|Ax\|_2 \leq (1 + \tau)\sqrt{n} \|x\|_2 \quad (x \in \mathbb{R}^d), \quad \tau := \frac{c_1 a_d}{10}. \quad (8)$$

For completeness, the exceptional probability is at most $2 \exp(-c\tau^2 n)$ once $\sqrt{d/n} \leq \tau/2$; here $a_d \geq c \log \log n / \log n$ and $d \leq c_0 \log n$, so it is smaller than 1/100 for $n \geq n_0$.

On the intersection of these events,

$$\frac{\|Au\|_p / \|Au\|_2}{\|Av\|_p / \|Av\|_2} \geq (1 + c_1 a_d) \frac{1 - \tau}{1 + \tau} > 1 + \frac{c_1}{2} a_d.$$

The left-hand side is bounded above by $D_p(A\mathbb{R}^d)$. The intersection has probability greater than $9/20 - 1/100 > 1/4$, proving the proposition with $\gamma = c_1/2$. $\square$

6 Full resolution

Theorem 5. *There are absolute constants C_0, C, n_0 such that the following holds. Let $n \geq n_0$, $p \geq C_0 \log n$, $0 < \varepsilon < 1/3$, and $k > 1$. If a Haar-random k -dimensional subspace E of $\mathbb{R}^n$ satisfies*

$$\mathbb{P}\{D_p(E) \leq 1 + \varepsilon\} > \frac{3}{4},$$

then

$$k \leq C \frac{\varepsilon \log n}{\log(1/\varepsilon)}. \quad (9)$$

Proof. First note a monotonicity fact. If E is $(1+\varepsilon)$ -spherical, every subspace of E is $(1+\varepsilon)$ -spherical. If one first chooses a Haar-random k -subspace and then a Haar-random d -subspace inside it, the marginal law of the latter is Haar on the d -dimensional Grassmannian. Hence the probability hypothesis in dimension k implies the same hypothesis in every dimension $2 \leq d \leq k$.

Put $L = \log n$ and

$$s = \frac{\varepsilon L}{\log(1/\varepsilon)}.$$

We first treat $0 < \varepsilon \leq \varepsilon_0$, where ε_0 is a sufficiently small absolute constant. Set

$$d = \max\{2, \lceil As \rceil\},$$

with A an absolute constant to be fixed. If $As \geq 1$, then $d \leq 2As$ and, after taking ε_0 small enough,

$$d \leq c_0 L, \quad \log(eL/d) \geq \log\left(\frac{e \log(1/\varepsilon)}{2A\varepsilon}\right) \geq \frac{1}{2} \log(1/\varepsilon).$$

Consequently

$$a_d = \frac{d \log(eL/d)}{L} \geq \frac{A}{2} \varepsilon. \quad (10)$$

If $As < 1$, then $d = 2$. The inequality $\varepsilon L / \log(1/\varepsilon) < 1/A$ implies, for large L ,

$$\varepsilon \leq \frac{C \log L}{AL}. \quad (11)$$

To see this, put $t = \log(1/\varepsilon)$. If $t \leq 2 \log L$, the assertion follows directly from $\varepsilon L < t/A$; if $t > 2 \log L$, then $\varepsilon < L^{-2}$, which is even smaller. Since $a_2 = 2 \log(eL/2)/L$, choosing A sufficiently large gives

$$\gamma a_2 > \varepsilon. \quad (12)$$

Choose A large enough for both (10) and (12) to imply $\gamma a_d > \varepsilon$. If $d \leq k$, Proposition 4 says that more than one quarter of the d -subspaces have distortion greater than $1 + \varepsilon$, contradicting the inherited probability hypothesis. Hence $k < d$. The case $As < 1$ is then impossible because $k \geq 2 = d$. In the remaining case, $k < d \leq 2As$, which is (9).

It remains to treat $\varepsilon_0 < \varepsilon < 1/3$. Here the elementary comparison

$$D_\infty(E) \leq n^{1/p} D_p(E)$$

has only a small absolute loss. Increase C_0 so that $e^{1/C_0} - 1 \ll \varepsilon_0$, and put $1 + \delta = (1 + \varepsilon)n^{1/p}$. Then $\delta < 1/2$ and $\delta \leq C\varepsilon$. Tikhomirov's sharp randomized cube theorem [2] gives

$$k \leq C_T \frac{\delta L}{\log(1/\delta)} \leq C \frac{\varepsilon L}{\log(1/\varepsilon)}.$$

This proves the theorem. $\square$

Remark 6 (Asymptotic formulation). *The source question is asymptotic in n , so Theorem 5 is stated for $n \geq n_0$. No parameter-dependent threshold is hidden: C_0, C, n_0 are absolute.*

7 Verification notes and reviewer focus

The proof is analytic and has no computational dependency. Its only external inputs are the standard Gaussian singular-value estimate and, for the fixed-distortion terminal case, Tikhomirov’s sharp cube theorem. The new small-distortion argument is contained in Lemmas 1–3 and Proposition 4.

The highest-priority human checks are:

1. verify the regular-conditional-law statement in Lemma 3 and the stochastic domination after choosing $v \perp u$;
2. check the uniformity in $2 \leq d \leq c_0 \log n$ of the Stirling calculation in (5);
3. check the inversion from a_d to $\varepsilon \log n / \log(1/\varepsilon)$, particularly the $As < 1$ cutoff;
4. confirm the $> 3/4$ versus $> 1/4$ probability complement and the Grassmannian dimension-monotonicity step.

An independent line-by-line audit is recorded in `verification_report.md`. The current verdict is **likely valid, candidate full resolution**, not human verified.

8 Novelty check and limitations

A bounded search on 9 August 2026 covered the run’s lightweight indexes, the local arXiv source corpus, exact title/keyword searches for randomized Dvoretzky in logarithmic p , the source DOI’s OpenAlex citation graph (24 citing records), and targeted arXiv/web searches combining “ ℓ_p ”, “ $\log(1/\varepsilon)$ ”, “logarithmic p ”, “ χ -distribution”, and the source title. Lytova and Tikhomirov still described the logarithmic randomized-Dvoretzky estimates as non-sharp in 2017 [3]. No later paper claiming this resolution, nor this conditional-longest-row soft-max argument, was found. This bounded search does not prove novelty.

The theorem resolves the source question in its asymptotic $p \gtrsim \log n$ meaning, with unspecified absolute constants. It does not optimize C_0 , C , the majority threshold, or the finite- n cutoff.

References

- [1] G. Paouris, P. Valettas and J. Zinn, *Random version of Dvoretzky’s theorem in ℓ_p^n* , Stochastic Process. Appl. **127** (2017), 3187–3227; arXiv:1510.07284.
- [2] K. E. Tikhomirov, *The randomized Dvoretzky’s theorem in ℓ_∞^n and the χ -distribution*, Geometric Aspects of Functional Analysis, Lecture Notes in Math. 2116, Springer (2014), 455–463.
- [3] A. Lytova and K. Tikhomirov, *The variance of the ℓ_p^n -norm of the Gaussian vector, and Dvoretzky’s theorem*, Algebra i Analiz **30** (2018), no. 4, 107–139; arXiv:1705.05052.
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